

SMART DIETER PROJECT

Health Assessment, Food Tracking & Personalized Recipe Recommendations

Executive Summary

Smart Dieter is an AI-powered nutrition management system that dynamically identifies food conflicts based on user health profiles, allergies, aversions, and dietary restrictions. The system flags unsafe meals and recommends personalized recipe alternatives from the BN recipe database.

1. System Overview

Smart Dieter operates on a three-phase implementation roadmap:

- Phase 1: Conflict Detection — Safety & Basic Restrictions
- Phase 2: Advanced Conflict Resolution — Expanded Constraints & Ingredient checklist Logic
- Phase 3: Intelligent Recipe Replacement — AI-Driven Personalized Recommendations

2. Phase 1: Conflict Detection

2.1 Overview

Phase 1 establishes the foundational safety layer. It ingests user health profiles and meal plans, then identifies meals containing ingredients that conflict with documented allergies or aversions.

2.2 Data Points Collected

Data Point	Description	Source
Eating Habits	User primary food patterns (Veg, Non-veg, Vegan)	User Profile
Allergies	Complete list of food allergies	Health Assessment
Aversions	Foods user dislikes or avoids	User Preferences
Recipes & Ingredients	Curated meals with ingredient lists	BN Database

2.3 Processing Workflow

Step 1: User Information Collection

1. User completes health assessment form
2. System stores: Allergies, Aversions, Eating Habits
3. User Profile created in secure database

Step 2: Diet Plan Parsing

4. Diet plan uploaded or selected from template
5. System extracts all dishes from the diet plan
6. Creates indexed meal list for processing

Step 3: Ingredient Extraction

7. For each dish, retrieve recipe from BN database
8. Extract complete ingredient list including sub-components
9. Normalize ingredient names for matching

Step 4: AI Conflict Detection

AI Prompt Rules:

- Read User Profile (Allergies, Aversions)
- Iterate through each Dish + Ingredient combination
- Identify exact ingredient matches against allergies
- Identify aversion matches
- Return: Flagged Dish + Conflicting Ingredient + Reason

Step 5: Dashboard Display

10. Conflicts displayed with RED highlighting
11. Each conflict shows: Dish Name, Ingredient, Conflict Type
12. User can acknowledge or request alternatives

2.4 Output & User Actions

Phase 1 Outputs:

- List of conflicting meals with reason
- Safe meal list (no conflicts detected)
- Conflict severity indicators

User Actions Available:

- Acknowledge conflict and proceed
- Remove meal from diet plan
- Request alternative

3. Phase 2: Advanced Conflict Resolution

3.1 Overview

Phase 2 expands conflict detection to include practical lifestyle constraints. It adds shopping limitations and cultural/religious dietary rules (e.g., Jain restrictions).

3.2 Additional Data Points

Data Point	Description	Example
Ingredient Can't Buy	Unavailable/unaffordable in region	Specialty spices
Jain Restrictions	Religious dietary rules	Onion, garlic, potatoes
Location	User's geographic location	India, Mumbai
Eating Habits	Veg/Non-veg/Vegan status	Vegetarian
Food Aversions	Extended preference list	Bitter vegetables

3.3 Enhanced AI Logic

The AI now checks for three types of conflicts:

Check 1: Health & Personal Restrictions

- Does dish contain user allergies?
- Does dish contain aversions?
- Does dish violate eating habit (veg/non-veg)?

Check 2: Cultural/Religious Restrictions

- Does dish contain restricted ingredients (Jain rules)?
- Severity: HIGH (often non-negotiable)

Check 3: Shopping Practicality

- Are all ingredients available in the user's region?
- Are all ingredients in the 'Can Buy' list?
- Flag unavailable ingredients separately

3.4 Dashboard Display

Conflict Type	Color	Action
Allergy	RED	High priority
Aversion	YELLOW	User choice
Cultural	ORANGE	Non-negotiable
Shopping	BLUE	Unavailable

4. Phase 3: Intelligent Recipe Replacement

4.1 Overview

Phase 3 automatically identifies the best replacement recipes. It uses AI to match replacements based on the user's eating habits, cuisine preferences, and meal frequency patterns.

4.2 Additional Data Points

Data Point	Description	Source
User Assessment (NAF)	Nutrition Assessment Framework	Health Assessment
Conflicting Dishes	Meals flagged in Phase 1 & 2	AI Output
BN Recipe Pool	100+ curated recipes per conflict	BN Database

4.3 Recipe Pool Generation

For each flagged dish, system generates 6 candidate recipes by:

13. Filter by Eating Habit: Veg/Non-veg/Vegan status
14. Filter by Cuisine Preference: User ethnicity or preferred cuisine
15. Filter by Meal Category: Match breakfast/lunch/dinner/snack
16. Filter by Food Frequency: User NAF preferences

4.4 Diet Combo Identification

The diet plan uses symbolic notation:

Symbol	Meaning	Example
OR	Choose one option	Rice OR Bread
/	Can be combined	Curry / Rice
+	Must serve together	Main Dish + Side

4.5 AI Recipe Selection

AI Input

- User Conflicting Dish
- BN Recipe Pool: 6 filtered candidates
- User Profile: NAF data
- Diet Combo: Symbolic notation (OR, /, +)
- Prompt Rules: AI decision criteria

AI Prompt Rules

17. Analyze user's NAF profile and eating preferences
18. Understand diet combo structure of original meal
19. Identify why original meal conflicted
20. From 6 candidates, rank by match quality

21. Return top 3 BN recipes as recommended alternatives

4.6 Output Display

For each conflict, dashboard displays:

- Original Conflicting Meal with reason
- Top 3 AI-Recommended Alternatives
- Why each alternative was recommended
- User ability to accept, modify, or view more options

5. System Architecture

5.1 Data Flow

User Data Collection → User Profile Storage → Diet Plan Input → Parse & Extract Dishes → Extract Ingredients from BN Database → AI Conflict Detection → Dashboard Display & User Actions → Updated Diet Plan

5.2 Key Components

Component	Function	Responsibility
User Data Module	Collect & store profile	Secure storage
Diet Parser	Extract dishes from meal plans	Data extraction
Recipe Database	Store recipes with ingredients	Maintain library
AI Engine	Detect conflicts & recommend	Phase 1/2/3 logic
Dashboard UI	Display results	User interface

6. Implementation Timeline

Phase	Duration	Deliverables
Phase 1	Weeks 1-3	Basic conflict detection, dashboard
Phase 2	Weeks 4-6	Enhanced logic, shopping rules
Phase 3	Weeks 7-10	AI recipe engine, recommendations

7. User Benefits

- Safety: Automatically identifies unsafe meals
- Personalization: Meals tailored to preferences
- Practicality: Accounts for regional ingredient availability
- Intelligence: AI suggests relevant alternatives
- Engagement: Users feel supported in dietary journey

8. Technical Stack

Backend: Node.js / Python with AI (Gemini API)

Database: MySQL (user profiles), MongoDB (recipes)

Frontend: React for dashboard

AI Engine: Gemini API

Vector Embeddings : Qdrant + MySQL

Hosting: AWS / GCP with secure data handling

5. Phase 4: Profile Mapping & Success-Based Diet Suggestions

5.1 Overview

Phase 4 leverages historical data. System maps current user to successful profiles, retrieves their proven diet, and refines it for immediate mentorship. Mentors review instead of building from scratch.

5.2 Data Points for Mapping

Demographics: Gender, Age, Country, State, City, Ethnicity, Weight

Health: Eating Habits, Allergies, Aversions, Restrictions, Food Frequency, Food Preferences (recall & frequency),

Program Status: COMPLETED profiles ONLY (exclude DROPOUT)

5.3 Processing Workflow

Step 1: Filter by Program Completion

22. Query database: SELECT profiles WHERE program_status = 'COMPLETED'

23. Exclude all dropouts (high confidence pool only)

Step 2: Demographic Filtering

24. Gender: EXACT match

25. Age: Range match (± 5 years)

26. Country: EXACT match

27. State/City: EXACT match (optional if <3 candidates)

28. Ethnicity: EXACT match (when available)

Step 3: Health Parameter Matching

29. Eating Habit: EXACT match (Veg/Non-veg/Vegan)
30. Allergies: Candidate must have NONE user is allergic to
31. Restrictions: Candidate restrictions $\subseteq$ user restrictions
32. Cuisine: Candidate cuisine aligns with user

Step 4: Success Ratio Calculation

For each matched profile, score: (Delta in Weight and Trackers)

- Diet Adherence: % meals followed (0-100)
- Goal Achievement: % health goals met (0-100)
- Completion Time: Speed to goal (normalized 0-100)

Success Ratio = (Adherence + Goals + Time) / 3

Sort by Success Ratio (highest first) → SELECT TOP profile

Step 5: Diet Extraction & Refinement

33. Retrieve FIRST diet plan from TOP matched profile
34. Run through Phase 1-3 conflict check (safety verification)
35. Flag any conflicts for mentor refinement
36. Present to mentor: Matched profile + Success ratio + refined diet

5.4 Mentor Workflow

37. Manual diet building: 30-45 mins → Phase 4 review: 5-10 mins
38. Review suggested diet + matched profile success data
39. Make minor tweaks if needed
40. Send immediately (no build time)

5.5 Dashboard Display

- Suggested Diet + Matched Profile Details
- Success Ratio: Why this diet works
- Conflict Check Results
- Alternative profiles (2-3 options)

5.6 Why It Works

- Diets from PROVEN users (not theoretical)
- Demographics match = cultural familiarity
- Health params match = similar challenges already solved
- Regional availability = less substitution needed
- Psychology: 'If they did it, so can I' confidence boost

6. System Architecture (Updated)

6.1 Complete Data Flow

User Assessment (1-3) → Phase 4 Profile Mapping → Matched Profile + Diet → Refine Diet by Running Phase 1-3 → Mentor Review → Send Diet → User

6.2 Phase 4 Database Requirements

- user_profiles: Demographics + health
- user_programs: Status (COMPLETED/DROPOUT)
- user_diet_plans: Diet per user
- success_metrics: Adherence, goals, time
- profile_matching_log: Track matches

7. Benefits

Users: Faster assignment, higher success rate, cultural familiarity, psychological confidence

Mentors: 75% time savings, evidence-backed diets, reduced failures, scalability

Organization: Better outcomes, faster delivery, lower mentor workload, data-driven insights

8. Technical Stack (Phase 4 Updates)

Backend: Profile matching engine + success ratio scoring

Database: PostgreSQL (profiles, programs), MongoDB (diets, metrics)

Frontend: Mentor interface for diet refinement + user dashboard

Analytics: Track profile matches + success outcomes

6. Clara Meal Plan: Macro-Based Diet from Existing Plans

6.1 Overview

Clara Meal Plan is paid subscription service. Uses existing diet templates + BN recipe database to create macro-balanced meal plans. Includes diet recall tracking. Bridges gap between leads and full Smart Dieter membership.

6.2 Clara Data Collection

Initial Assessment:

- Age, Gender, Height, Weight, Location
- Eating Habit (Veg/Non-veg/Vegan), Allergies, Aversions
- Health Goal (weight loss, muscle, maintenance, energy)
- Activity Level (sedentary, moderate, active)

Diet Recall (3-day snapshot using Nutriscan):

- What user ate last 3 days (meal photos or text)
- Nutriscan parses meals → estimates macros (protein, carbs, fat) and records it.
- Baseline macro intake identified

6.3 Macro Calculation

Based on: Age, Weight, Activity, Goal

- TDEE (Total Daily Energy Expenditure) calculated
- Protein target: 1.2-2.2g per kg (goal dependent)
- Carb/Fat split: Based on activity + preference
- Example: 75kg user, weight loss = 1800 cal, 135g protein, 180g carb, 60g fat

6.4 Diet Template Selection

System filters existing diets by:

41. Eating Habit match (Veg/Non-veg/Vegan)
42. Location/Cuisine match (regional availability)
43. Macro profile match (closest to calculated target)
44. Select TOP 3 existing diets as base templates

6.5 Recipe Substitution & Customization

From TOP template, apply user constraints:

45. Phase 1 logic: Remove allergies/aversions
46. Recipe swap: Replace flagged meals with BN recipes matching:
 - Same macros ($\pm 10\%$)
 - Same meal category (breakfast/lunch/dinner)
 - User location + cuisine
47. Build final 7-day macro-balanced meal plan

6.6 Clara Workflow

Step 1: Lead Signup → Assessment

48. Lead pays subscription fee (₹X/month or ₹XX/quarter)
49. Complete onboarding form (demographics + health)

Step 2: Diet Recall

50. Upload photos or list meals from last 3 days
51. Clara AI (Gemini) parses: Ingredients, portions, macros
52. Baseline macro profile established

Step 3: Macro Target Calculation

53. TDEE calculated from: Age, weight, activity, goal
54. Protein, carb, fat targets set

Step 4: Template Selection

55. Filter existing diets: Eating habit, location, macros

56. Select best matching template

Step 5: Recipe Customization

- 57. Apply allergies/aversions (Phase 1)
- 58. Swap conflicting recipes from BN database
- 59. Match by: Macros $\pm 10\%$, meal type, location
- 60. Generate final 7-day plan

Step 6: Delivery

- 61. User gets: 7-day meals, recipes, shopping list, macro breakdown
- 62. Log meals daily in Clara app
- 63. Track macros automatically

6.7 Clara Features

Subscriber gets:

- Personalized macro-based meal plan (weekly or monthly)
- Shopping list (auto-generated from recipes)
- Meal logging + macro tracking (daily)
- Recipe substitutions (if user wants variety)
- Progress dashboard (weight, macros, adherence)
- Email weekly check-ins (no mentor, auto-generated)

6.8 Clara vs Smart Dieter vs Clara Premium

Feature	Clara	Clara + Mentor	Smart Dieter
Price	₹299/mo	₹799/mo	₹1499/mo
Meal Plan	AI macro-based	AI macro-based	Mentor + AI
Diet Recall	Initial only	Initial + monthly	Complete profile
Phase 4 Match	No	No	Yes (proven)
Mentor Support	No	App Chat	App chat + calls
Personalization	Good	Good	Excellent

6.9 Clara + Smart Dieter Upgrade Path

Lead journey:

- 64. Try Clara ₹299/mo (macro plan, no mentor)
- 65. After 3 months: 'Want results? Upgrade to Clara + Mentor'
- 66. After 6 months: 'Ready for full transformation? Smart Dieter (Phase 4)'

67. Customer lifetime: Clara → Clara+Mentor → Smart Dieter

6.10 Clara Backend

- Existing diet database: Filter by macro + location
- BN recipe database: Swap recipes by macro match
- Gemini API: Parse diet recall + extract macros
- Phase 1 engine: Remove allergies/aversions
- Macro calculator: TDEE + protein/carb/fat targets
- Meal logger: Track daily intake + generate reports

6.11 Clara Positioning

Marketing message:

- 'Personalized macro meal plan for ₹299/mo. No mentoring needed if you're self-disciplined.'
- 'AI builds your diet from your eating patterns.'
- 'Full macro tracking. Weekly updates. Simple.'
- 'Need help? Upgrade to mentor for ₹800/mo.'

7. System Architecture (Complete)

7.1 Three Product Tiers

Tier 1 - Clara (₹299/mo): AI macro plan, no mentor

Tier 2 - Clara + Mentor (₹799/mo): AI plan + email mentor

Tier 3 - Smart Dieter (₹1499/mo): Phase 4 (proven diets) + full mentor

7.2 Shared Infrastructure

All tiers use:

- Phase 1-3: Conflict detection + recipe alternatives
- BN recipe database: Shared recipes
- Gemini API: AI logic for personalization
- User profile: Progressive data collection

8. Conclusion

Smart Dieter ecosystem complete. Clara = affordability + simplicity. Clara+Mentor = affordability + support. Smart Dieter = proven + premium + transformation. Same tech (Phases 1-3), different value. Multiple revenue streams. Customer can upgrade as needs grow.